11. Which of the following is an electromagnetic wave?
    (a) α - rays (b) β - rays
    (c) γ - rays (d) all of them

12. Which one of them is used to produce a propagating electromagnetic wave?.
    (a) an accelerating charge
    (b) a charge moving at constant velocity
    (c) a stationary charge
    (d) an uncharged particle

13. Let $E = E_o \sin[10^6 x - \omega t]$ be the electric field of plane electromagnetic wave, the value of ω is
    (a) 0.3×10^{-14} rad s^{-1}
    (b) 3×10^{-14} rad s^{-1}
    (c) 0.3×10^{14} rad s^{-1}
    (d) 3×10^{14} rad s^{-1}

14. Which of the following is NOT true for electromagnetic waves?.
    (a) it transport energy
    (b) it transport momentum
    (c) it transport angular momentum
    (d) in vacuum, it travels with different speeds which depend on their frequency

15. The electric and magnetic fields of an electromagnetic wave are
    (a) in phase and perpendicular to each other
    (b) out of phase and not perpendicular to each other
    (c) in phase and not perpendicular to each other
    (d) out of phase and perpendicular to each other

Answers

1) b	**2)** d	**3)** d	**4)** c	**5)** a
6) b	**7)** a	**8)** c	**9)** b	**10)** b
11) c	**12)** a	**13)** d	**14)** d	**15)** a

II Short answer questions

1. What is displacement current?
2. What are electromagnetic waves?
3. Write down the integral form of modified Ampere's circuital law.
4. Explain the concept of intensity of electromagnetic waves.
5. What is meant by Fraunhofer lines?

III Long answer questions

1. Write down Maxwell equations in integral form.
2. Write short notes on (a) microwave (b) X-ray (c) radio waves (d) visible spectrum
3. Discuss briefly the experiment conducted by Hertz to produce and detect electromagnetic spectrum.
4. Explain the Maxwell's modification of Ampere's circuital law.
5. Write down the properties of electromagnetic waves.
6. Discuss the source of electromagnetic waves.
7. What is emission spectra?. Give their types.
8. What is absorption spectra?. Give their types.

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